```
name = [1,2,3,4]



for i in name:
    print(i)
```

Show hidden output

```
# # ============================================================
# #                    PYTHON FOR LOOP
# # ============================================================

# # WHY DO WE NEED A FOR LOOP?
# #
# # Suppose we want to print "Hello" 5 times.
# #
# # Without a loop:

# print("Hello")
# print("Hello")
# print("Hello")
# print("Hello")
# print("Hello")

# # This works...
# #
# # But what if we want to print "Hello"
# # 1000 times?
# #
# # Writing print() 1000 times is not practical.
# #
# # To avoid repeating the same code,
# # Python provides LOOPS.
# #
# # A loop repeats a block of code.





# # ============================================================
# #                    FOR LOOP
# # ============================================================

# # A FOR loop is used when we already know
# # how many times we want to repeat something.
# #
# # Syntax:
# #
# # for variable in sequence:
# #     code


# # ============================================================
# # Example 1
# # ============================================================

# for i in range(5):
#     print(i)

# # Output:
# #
# # 0
# # 1
# # 2
# # 3
# # 4

# # Explanation:
# #
# # range(5) generates:
# #
# # 0, 1, 2, 3, 4
# #
# # One value is taken at a time.
# #
# # Iteration 1
# # i = 0
```

```
# #
# # Iteration 2
# # i = 1
# #
# # Iteration 3
# # i = 2
# #
# # Iteration 4
# # i = 3
# #
# # Iteration 5
# # i = 4
# #
# # Then the loop ends.


# # ============================================================
# # Example 2
# # range(start, stop)
# # ============================================================

# for i in range(1, 6):
#     print(i)

# # Output:
# #
# # 1
# # 2
# # 3
# # 4
# # 5

# # Explanation:
# #
# # start = Included
# #
# # stop = Excluded
# #
# # range(1,6)
# #
# # means
# #
# # 1,2,3,4,5


# # ============================================================
# # Example 3
# # range(start, stop, step)
# # ============================================================

# for i in range(2, 11, 2):
#     print(i)

# # Output:
# #
# # 2
# # 4
# # 6
# # 8
# # 10

# # Explanation:
# #
# # Start = 2
# #
# # Stop = 11 (11 is not included)
# #
# # Step = 2
# #
# # So Python jumps by 2 each time.


# # ============================================================
# # Example 4
# # Printing a message multiple times
# # ============================================================

# for i in range(5):
#     print("Welcome")

# # Output:
# #
# # Welcome
```

```
# # Welcome
# # Welcome
# # Welcome
# # Welcome


# # ============================================================
# # Example 5
# # Looping through a List
# # ============================================================

# fruits = ["Apple", "Banana", "Mango"]

# for fruit in fruits:
#     print(fruit)

# # Output:
# #
# # Apple
# # Banana
# # Mango

# # Explanation:
# #
# # fruit is just a variable.
# #
# # Python takes one element from the list
# # during each iteration.
# #
# # Iteration 1
# # fruit = Apple
# #
# # Iteration 2
# # fruit = Banana
# #
# # Iteration 3
# # fruit = Mango


# # ============================================================
# # Example 6
# # Looping through a Tuple
# # ============================================================

# days = ("Monday", "Tuesday", "Wednesday")

# for day in days:
#     print(day)

# # Tuples can also be iterated.


# # ============================================================
# # Example 7
# # Looping through a Set
# # ============================================================

# numbers = {10, 5 , 20, 30}

# for number in numbers:
#     print(number)

# # NOTE:
# #
# # Sets are unordered.
# #
# # So the output order may change.


# # ============================================================
# # Example 8
# # Looping through a Dictionary
# # ============================================================

# student = {
#     "name": "John",
#     "age": 21,
#     "course": "Data Science"
# }

# for key in student:
#     print(key)
```

```
# # Output:
# #
# # name
# # age
# # course

# # To print both key and value:

# for key, value in student.items():   -> (name:john, )
#     print(key, ":", value)

# # Output:
# #
# # name : John
# # age : 21
# # course : Data Science


# # ============================================================
# # Example 9
# # Looping through a String
# # ============================================================

# name = "Python"

# for letter in name:
#     print(letter)

# # Output:
# #
# # P
# # y
# # t
# # h
# # o
# # n


# # ============================================================
# # PRACTICE QUESTIONS
# # ============================================================

# # 1. Print numbers from 1 to 10.


# for i in range(1,11):
#   print(i)

# # 2. Print numbers from 10 to 1.

# for num in range(10, 0, -1):  # start=10, stop=0 (exclusive), step=-1
#     print(num)

# # 3. Print even numbers from 2 to 20.

# for i in range(21):  #i=3
#   if i%2== 0:
#     print(i)


# # 4. Print odd numbers from 1 to 19.

# # 5. Print your name 10 times.

# for name in range(10):
#   print("ganesh")

# # 6. Print all fruits from a list.

# # 7. Print all characters in your name.

name = "ganesh"

for i in name:
  print(i)

# # 8. Print every key in a dictionary.

# # 9. Print every value in a dictionary.

# # 10. Print the multiplication table of 5.
```

```
for i in range(1,11):
  print(f"5 * {i} = {5*i}")


# # ============================================================
# # SUMMARY
# # ============================================================

# # A FOR LOOP is used when
# # the number of repetitions is known.
# #
# # It can iterate over:
# #
# # ✓ range()
# # ✓ List
# # ✓ Tuple
# # ✓ Set
# # ✓ Dictionary
# # ✓ String
# #
# # Easy way to remember:
# #
# # "FOR = I know how many times
# # or I have a collection to iterate through."
```

```
g
a
n
e
s
h
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

```
word = input("enter the word ! ")  # google

count = 0

for i in word:
  count += 1

print(count)
```

```
enter the word ! google
6
```

```
age = []

for i in range(5):
  age.append(int(input("enter your age:")))

# print(age)

 # normal way of writing list adn for loop

# a2
```

```
enter your age:21
enter your age:22
enter your age:3
enter your age:44
enter your age:22
```

```
# sq = []

# for i in range(1,11):
#   sq.append(i*i)
# print(sq)

# # new_list = [expression for variable in iterable   if else condition]
```

```
sq = [i*i for i in range(1,11) ]
sq
```

```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```

```
even_numbers = [i for i in range(1, 21) if i % 2 == 0]

even_numbers

# even_num = []

# for i in range(1,21):
#   if i%2 == 0:
#     even_num.append(i)
```

```
[2, 4, 6, 8, 10, 12, 14, 16, 18, 20]
```

```
name = "ganesh"

name.upper()

name.lower()
```

```
'ganesh'
```

```
names = ["john", "alice", "bob"]
# upper_case = []

# for i in names:
#   upper_case.append(i.upper())




upper_case = [ i.upper() for i in names]
upper_case
```

```
['JOHN', 'ALICE', 'BOB']
```

```
2# ============================================================
#              PYTHON LIST COMPREHENSIONS
# ============================================================

# WHAT IS A LIST COMPREHENSION?

# list and for loop .
#
# A List Comprehension is a SHORTER and CLEANER way
# to create a new list.
#
# Instead of writing multiple lines using a for loop,
# we can write everything in a single line.


# ============================================================
# Example 1
# Creating a List using a FOR LOOP
# ============================================================

numbers = []

for i in range(1, 6):
    numbers.append(i)

print(numbers)

# Output
# [1, 2, 3, 4, 5]

# Here,
#
# Step 1
# Create an empty list.
#
# Step 2
# Run a for loop.
#
```

```
# Step 3
# Append each value.
#
# This works perfectly,
# but Python gives us a shorter way.


# ============================================================
# Example 2
# Using List Comprehension
# ============================================================

numbers = [ i for i in range(1, 6)]

print(numbers)

# Output
# [1, 2, 3, 4, 5]

# This does exactly the same thing
# in just ONE line.


# ============================================================
# General Syntax
# ============================================================

# new_list = [expression for variable in iterable]
numbers = [ i for i in range(6)]

#
# expression  -> What should be stored?
#
# variable    -> Current item
#
# iterable    -> List, Tuple, String, range(), etc.


# ============================================================
# Example 3
# Squares of Numbers
# ============================================================
squares = [1,4,]

for i in range(1,6):
  squares.append(i*i)




squares = [i * i for i in range(1, 6)]

print(squares)

# Output
# [1, 4, 9, 16, 25]


# ============================================================
# Example 4
# Even Numbers
# ============================================================

even_numbers = [i for i in range(1, 21) if i % 2 == 0 ]

print(even_numbers)

# Output
# [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]

# if condition filters the values.


# ============================================================
# Example 5
# Odd Numbers
# ============================================================

odd_numbers = [i for i in range(1, 21) if i % 2 != 0]

print(odd_numbers)
```

```python
# Output
# [1, 3, 5, 7, 9, 11, 13, 15, 17, 19]


# ============================================================
# Example 6
# Multiplication Table of 5
# ============================================================

table = [5 * i for i in range(1, 11)]

print(table)

# Output
# [5, 10, 15, 20, 25, 30, 35, 40, 45, 50]


# ============================================================
# Example 7
# Convert Names to Uppercase
# ============================================================

names = ["john", "alice", "bob"]

upper_case = [JOHN,]

for i in names:
  upper_case.append(i.upper()# JOHN)


uppercase = [name.upper() for name in names]

print(uppercase)

# Output
# ['JOHN', 'ALICE', 'BOB']


# ============================================================
# Example 8
# Length of Each Word
# ============================================================

fruits = ["Apple", "Banana", "Mango"]

lengths = [len(fruit) for fruit in fruits]

print(lengths)

# Output
# [5, 6, 5]


# ============================================================
# Example 9
# Filter Long Words
# ============================================================

words = ["cat", "elephant", "dog", "python"]

long_words = [word for word in words if len(word) > 3]

print(long_words)

# Output
# ['elephant', 'python']


# ============================================================
# Example 10
# First Letter of Every Word
# ============================================================

names = ["Alice", "Bob", "Charlie"]

letters = [name[0] for name in names]

print(letters)

# Output
# ['A', 'B', 'C']
```

```python
# ============================================================
# FOR LOOP vs LIST COMPREHENSION
# ============================================================

# FOR LOOP

numbers = []

for i in range(1, 6):
    numbers.append(i)

print(numbers)


# LIST COMPREHENSION

numbers = [i for i in range(1, 6)]

print(numbers)

# Both produce:
#
# [1, 2, 3, 4, 5]
#
# List Comprehension is simply
# shorter and easier to read.


# ============================================================
# PRACTICE QUESTIONS
# ============================================================

# 1. Create a list from 1 to 20.

# 2. Create a list of squares from 1 to 10.

# 3. Create a list of cubes from 1 to 10.

# 4. Create a list of even numbers from 1 to 50.

# 5. Create a list of odd numbers from 1 to 50.

# 6. Convert all names to uppercase.

# 7. Find the length of every word in a list.

# 8. Create a list of numbers divisible by 5.

# 9. Extract the first letter from every name.

# 10. Filter numbers greater than 50.


# ============================================================
# SUMMARY
# ============================================================

# FOR LOOP
#
# Used to perform an action repeatedly.
#
# Example:
#
# for i in range(5):
#     print(i)


# LIST COMPREHENSION
#
# Used to CREATE A NEW LIST
# in a short and clean way.
#
# General Syntax:
#
# [expression for variable in iterable]
#
# With Condition:
#
# [expression for variable in iterable if condition]
```

```
# for i in range(1,11):
#   if i%2 == 0:
#     print(i)

number = 1
while number <=10:
  if number%2 == 0:
    print(number)
  number = number + 1
```

```
2
4
6
8
10
```

```
number = 2

while number <= 20:
  number += 2
  print(number)
```

```
4
6
8
10
12
14
16
18
20
22
```

```
count = 10

while count >= 1:
    print(count)
    count -= 1
print("Time's Up!")
```

```
10
9
8
7
6
5
4
3
2
1
Time's Up!
```

```
password = "anans"

while password:
  password = input("enter the password: ")
  if password == "anans":
    break


print("access granted")
```

```
enter the password: saka
enter the password: asmkas
enter the password: amsk
enter the password: anans
access granted
```

```
number = int(input("Enter a number: "))

while number != 0:
    print("You entered:", number)
    number = int(input("Enter another number (0 to stop): "))

print("Loop Ended.")
```

```
Enter a number: 10
You entered: 10
Enter another number (0 to stop): 0
Loop Ended.
```

```
while True:
  pass
```

```
---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call last)
/tmp/ipykernel_2557/994876226.py in <cell line: 0>()
----> 1 while True:
      2   pass

KeyboardInterrupt:
```

```
# ============================================================
#                  PYTHON WHILE LOOP
# ============================================================

# WHY DO WE NEED A WHILE LOOP?
#
# Sometimes we DO NOT know how many times
# a loop should run.
#
# Example:
#
# Keep asking the user for the correct password.
#
# We don't know if the user will enter
# the correct password on the 1st attempt,
# 2nd attempt, or even the 10th attempt.
#
# In such cases, we use a WHILE LOOP.


# ============================================================
#                  WHILE LOOP SYNTAX
# ============================================================

# while condition:
#     code




# ============================================================
# Example 1
# Print Numbers from 1 to 5
# ============================================================

for i in range(1,6):
  print(i)

count = 1

while count <= 5:
    print(count)
    count += 1

# Output
#
# 1
# 2
# 3
# 4
# 5

# Explanation
#
# count = 1
#
# Is 1 <= 5 ?
# Yes
#
# Print 1
#
# Increase count by 1
```

```
#
# Repeat until count becomes 6.
#
# When count = 6
#
# Is 6 <= 5 ?
#
# No
#
# Loop Stops.


# ============================================================
# Example 2
# Print "Hello" 5 Times
# ============================================================

for i in range(5):
  print(
    "hello"
  )

count = 1
while count <= 5:
    print("Hello")




# ============================================================
# Example 3
# Print Even Numbers
# ============================================================

number = 2

while number <= 20:
  number += 2


# ============================================================
# Example 4
# Countdown Timer
# ============================================================

count = 10

while count >= 1:
    print(count)
    count -= 1

print("Time's Up!")


# ============================================================
# Example 5
# User Input
# ============================================================

number = int(input("Enter a number: "))

while number != 0:
    print("You entered:", number)

    number = int(input("Enter another number (0 to stop): "))

print("Loop Ended.")

# Explanation
#
# Continue asking the user for numbers
# until they enter 0.


# ============================================================
# Example 6
# Password Checker
# ============================================================

password = ""

while password != "python123":
```

```python
    password = input("Enter Password: ")  #abc@123

print("Access Granted!")

# The loop continues
# until the correct password is entered.


# ============================================================
# Infinite Loop
# ============================================================

# Be careful!
#
# If the condition never becomes False,
# the loop never stops.

# Example (DO NOT RUN)

# count = 1
#
# while count <= 5:
#     print(count)

# Why?
#
# count is never increased.
#
# So,
#
# count <= 5
#
# is always True.
#
# This is called an INFINITE LOOP.


# ============================================================
# BREAK, continue ,  pass
# ============================================================

# break immediately exits the loop.

count = 1

while True:
    print(count)

    if count == 101:
        break

    count += 1

print("Loop Finished")

# Output
#
# 1
# 2
# 3
# 4
# 5
# Loop Finished


# ============================================================
# CONTINUE
# ============================================================

# continue skips the current iteration.

count = 0

while count < 5:

    count += 1

    if count == 3:
        continue

    print(count)   #1 ,2,4,5
```

```
# Output
#
# 1
# 2
# 4
# 5

# Number 3 is skipped.


# ============================================================
# FOR LOOP vs WHILE LOOP
# ============================================================

# FOR LOOP
#
# ✓ Use when you know
# how many times to repeat.
#
# Example:
#
# Print numbers from 1 to 10.


# WHILE LOOP
#
# ✓ Use when you know
# the stopping condition.
#
# Example:
#
# Keep asking for a password
# until it is correct.


# ============================================================
# PRACTICE QUESTIONS
# ============================================================

# 1. Print numbers from 1 to 20.

# 2. Print numbers from 20 to 1.

count = 20

while count>=1:
  print(count)
  count = count - 1
# 3. Print even numbers from 2 to 50.

# 4. Print odd numbers from 1 to 49.

# 5. Print your name 10 times.

# 6. Create a countdown from 10 to 1.

# 7. Ask the user to enter numbers until they enter 0.

# 8. Keep asking for a password until it is correct.

# 9. Print the multiplication table of 8.

# 10. Print all numbers divisible by 5 from 1 to 100.


# ============================================================
# SUMMARY
# ============================================================

# FOR LOOP
#
# Use when the number of repetitions
# is known.
#
# Example:
#
# for i in range(5):
#     print(i)


# WHILE LOOP
#
```

```
# Use when the number of repetitions
# is NOT known.
#
# Example:
#
# while password != "python123":
#     password = input("Enter Password: ")


# Easy way to remember:
#
# FOR    -> I know HOW MANY times.
#
# WHILE  -> I know WHEN TO STOP.
```

```
  File "/tmp/ipykernel_2557/3154183459.py", line 2
    as
    ^
SyntaxError: invalid syntax
```